

Program : Diploma in Polymer Technology	
Course Code : 3094	Course Title: Synthetic Rubbers
Semester : 3	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3,T:1,P:0)	Periods per semester: 60

Course Objectives:

- To explain the structure preparation, properties and applications of various general and special purpose elastomers
- To explain the structure preparation, properties and applications of various speciality synthetic rubbers, thermoplastic elastomers and blends

Course Prerequisites:

Topic	Course code	Course name	Semester
Monomer preparation of synthetic rubbers		Fundamentals of polymer science	2

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive level
CO1	Explain the preparation, properties, compounding, curing and applications of general purpose synthetic rubbers.	18	Understanding
CO2	Explain preparation, properties, compounding, curing and applications of special purpose synthetic rubbers.	16	Understanding
CO3	Describe the properties, compounding, curing and applications of specialty synthetic rubbers.	12	Understanding
CO4	Describe the preparation, properties and applications of thermoplastic elastomers and blends	12	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	Explain the preparation, properties, compounding, curing and applications of general purpose synthetic rubbers.		
M1.01	Describe manufacturing of the following synthetic rubbers; SBR, BR, IR, IIR and EPDM.	6	Understanding
M1.02	Generalize the compounding of SBR, BR, IR, IIR and EPDM.	5	Understanding
M1.03	Generalize the curing of SBR, BR, IR, IIR and EPDM.	3	Understanding
M1.04	Chart out the properties and applications of SBR, BR, IR, IIR and EPDM.	4	Understanding
Contents: General purpose rubber-SBR, BR, IR, IIR EPDM–Manufacturing, properties, compounding, curing, and applications.			
CO2	Explain preparation, properties, compounding, curing and applications of special purpose synthetic rubbers.		
M2.01	Describe the manufacturing of the following synthetic rubber NBR, CR, CSM and silicone rubber	6	Understanding
M2.02	Generalize the compounding of NBR, CR, CSM and silicone rubber.	3	Understanding
M2.03	Generalize the curing of NBR, CR, CSM and silicone rubber.	3	Understanding
M2.04	Chart out the properties and applications of NBR, CR, CSM and silicone rubber.	4	Understanding
	Series Test – I	1	
Contents:			

Special purpose synthetic rubbers-NBR, CR, CSM, Silicone rubber-Manufacturing, compounding, curing, vulcanisate properties and applications- NBR, CR, CSM and silicone

CO3	Describe the properties, compounding, curing and applications of specialty synthetic rubbers.		
M3.01	Identify the properties and applications of Acrylic, Fluorocarbon and halobutyls	4	Understanding
M3.02	Generalize the compounding and curing of specialty rubbers such as Acrylic, Fluorocarbon, halobutyls	4	Understanding
M3.03	Identify the properties and applications of specialty synthetic rubber Carboxylated NBR, HTPB and HNBR.	4	Understanding

Contents:

Specialty synthetic rubbers- Properties, compounding, curing, and applications of Acrylic, Fluorocarbon, halobutyls -Properties and applications of specialty synthetic rubber Carboxylated NBR (XNBR), HTPB and HNBR.

CO4	Describe the preparation, properties and applications of thermoplastic elastomers and blends		
M4.01	Define thermoplastic elastomers , list out advantages and disadvantages of TPE	3	Understanding
M4.02	Explain the properties and application of the following thermoplastic elastomers S-I-S, S-B-S, EVA and TPU.	3	Understanding
M4.03	Generalize elastomer blends and blending techniques.	3	Understanding
M4.04	Explain the properties and application of the following blends NR/BR, NR/SBR, NR/EVA and NBR/PVC.	3	Understanding
	Series Test – II	1	

Contents:

TPE -Definition advantages and disadvantages. Properties and application of S-I-S, S-B-S, EVA,

TPU. Polymer blends- definition and blending techniques. Properties and applications of NR/BR, NR/SBR, NR/BR/SBR and NR/EVA

Text / Reference:

T/R	Book Title/Author
T1	Synthetic rubbers- D.C.Blackley (Publisher – Springer Netherlands Edition 1)
R1	Rubber Technology - Mauris Morton(Publisher – Springer Netherlands 1999 Edition)
R2	Vanderbilt Rubber hand book - R.T Vanderbilt (Publisher- R.T Vanderbilt 1990 Edition)
R3	Rubber Technology and Manufacture- C.M. Blow(Publisher- Butterworth Hainemann Limited, 1982 Edition)
R4	Training Manual - Rubber Training Institute, RRII, Kottayam
	Rubbery Materials and their compounds - J.A Brydson(Publisher –Elsevier Applied Science, London, 1988)

Online Resources:

Sl.No	Website Link
1	http://www.industrialrubbergoods.com
2	https://en.wikipedia.org/wiki/Synthetic_rubber
3	https://www.sciencedirect.com/
4	http://www.phoenix-compounding.com
5	https://iisrp.com/synthetic-rubber